



**MALLA REDDY TECHNICAL CAMPUS**  
MAISAMMAGUDA, MEDCHAL, HYDERABAD-500100  
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING  
**B.TECH III YEAR II SEMESTER REGULAR EXAMINATIONS**

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|-------------------------------------------------|--------------------------|----------------------------|
| Branch / Dept: Computer Science and Engineering | Course Code: R20-CSE     | Subject Code: CS601PC      |
| Subject: Design and Analysis of Algorithms      | Date of Exam: 15-11-2026 | Max Marks: 25 Marks        |
| Semester / Year: III Year II Semester           | Academic Year: 2026-2027 | Time / Duration: 1:30Hours |

**General Instructions:**

- 1. Answer all questions in Part A.
- 2. In Part B, answer either (a) or (b) from each question.
- 3. Assume suitable missing data if necessary.

**SECTION I**

**SHORT ANSWERS**

**Instructions:** Answer all 5 questions. [ 5 Questions × 2M | Attempt any 5 | Max Marks: 10 ]

1. Illustrate the step-by-step mechanism of Introduction: Algorithm using a concrete example. [2 Marks]
2. Discuss how Disjoint Sets: Disjoint set operations handles edge cases, boundary conditions, or exceptions. [2 Marks]
3. Explain the fundamental working principle of Dynamic Programming: General method with suitable illustrations. [2 Marks]
4. Illustrate the step-by-step mechanism of Greedy method: General method using a concrete example. [2 Marks]
5. Illustrate the step-by-step mechanism of Branch and Bound: General method using a concrete example. [2 Marks]

**LONG ANSWERS**

**Instructions:** Answer any 3 questions out of 5. [ 5 Questions × 5M | Attempt any 3 | Max Marks: 15 ]

6. Explain the fundamental working principle of Performance Analysis-Space complexity with suitable illustrations. [5 Marks]
7. Illustrate the step-by-step mechanism of union and find algorithms using a concrete example. [5 Marks]
8. How is applications- Optimal binary search tree implemented in modern software systems? Explain with pseudocode or schema. [5 Marks]
9. Explain the fundamental working principle of applications- Job sequencing with deadlines with suitable illustrations. [5 Marks]
10. How is applications implemented in modern software systems? Explain with pseudocode or schema. [5 Marks]